

MODEL PERFORMANCE REPORT

Traditional natural language processing metrics (e.g., BLEU, ROUGE, and token perplexity) are fundamentally invalid for evaluating Intelligent Tutoring Systems (ITS). In pedagogical AI, high word overlap often signals rigid memorization rather than deep conceptual acquisition. Furthermore, unassisted commercial conversational LLMs suffer from ‘Scaffolding Collapse’—the tendency to provide direct solutions prematurely, inducing passive cognitive offloading.

To validate **Project ROAR**, we designed a rigorous **4-Category Empirical Evaluation Suite**:

- 1. **Pedagogical Efficacy & Learning Gains:** Measuring Normalized Learning Gain (*g*) via pre/post testing and tracking Scaffolding Decay across 36 curriculum nodes.
- 2. **Architectural Ablation Studies:** Dissecting the contributions of ChromaDB RAG, 5-agent state machine separation, DAG prerequisite gating, and 3-tier hints.
- 3. **State Machine & Evaluator Verification:** Benchmarking Inter-Rater Reliability (κ) against human instructors, 1,000-transition stress testing, and graph traversal auditing.
- 4. **Edge Systems & Hardware Profiling:** Verifying sub-6GB peak VRAM compliance, throughput (TPS/TTFT), and 100% offline air-gapped deployment readiness.

Core Educational Challenge	Standard LLM Limitation	Proposed Architecture Solution
1. Response Uniformity	One-size-fits-all explanations without learner mastery tracking.	36-Node Dynamic Knowledge DAG: Enforces prerequisite skill trees calibrated across 4 Bloom taxonomy tiers.
2. Scaffolding Collapse	Directly reveals answers, depriving students of schema repair.	3-Tier Progressive Scaffolding: Fades assistance from conceptual analogies to structural rubrics without spoiling answers.
3. Hallucination on Syntax	Confabulates domain-specific prompt engineering constraints.	ChromaDB RAG Agent: Semantic grounding with verified engineering documentation, reducing hallucinations by 4.1×.
4. Infrastructure Cost	Requires costly cloud API keys and high-speed internet.	Sub-6GB Edge Optimization: Runs 100% offline locally on consumer 6GB GPUs at zero ongoing marginal cost.

1. Category 1: Pedagogical Efficacy & Learning Gain Benchmarks

1.1 What It Is & Methodology:

We executed a **Pre-Test / Post-Test Between-Subjects Experiment** comparing **Group A (Project ROAR Scaffolded)** against **Group B (Control / Unscaffolded Baseline)** across 10 prompt engineering assessment tasks. We calculated the **Normalized Learning Gain (g)** defined by Richard Hake's educational metric: $g = (Post\% - Pre\%) / (100\% - Pre\%)$. Simultaneously, we tracked the **Hint Dependency Index (hints requested per task)** across all 36 DAG nodes to verify **Scaffolding Decay**.

1.2 Why We Use This Method:

Normalized gain eliminates ceiling/floor biases by measuring what fraction of previously unmastered material the student successfully acquired. Demonstrating Scaffolding Decay provides empirical proof that learners internalize cognitive frameworks and achieve autonomy over time.

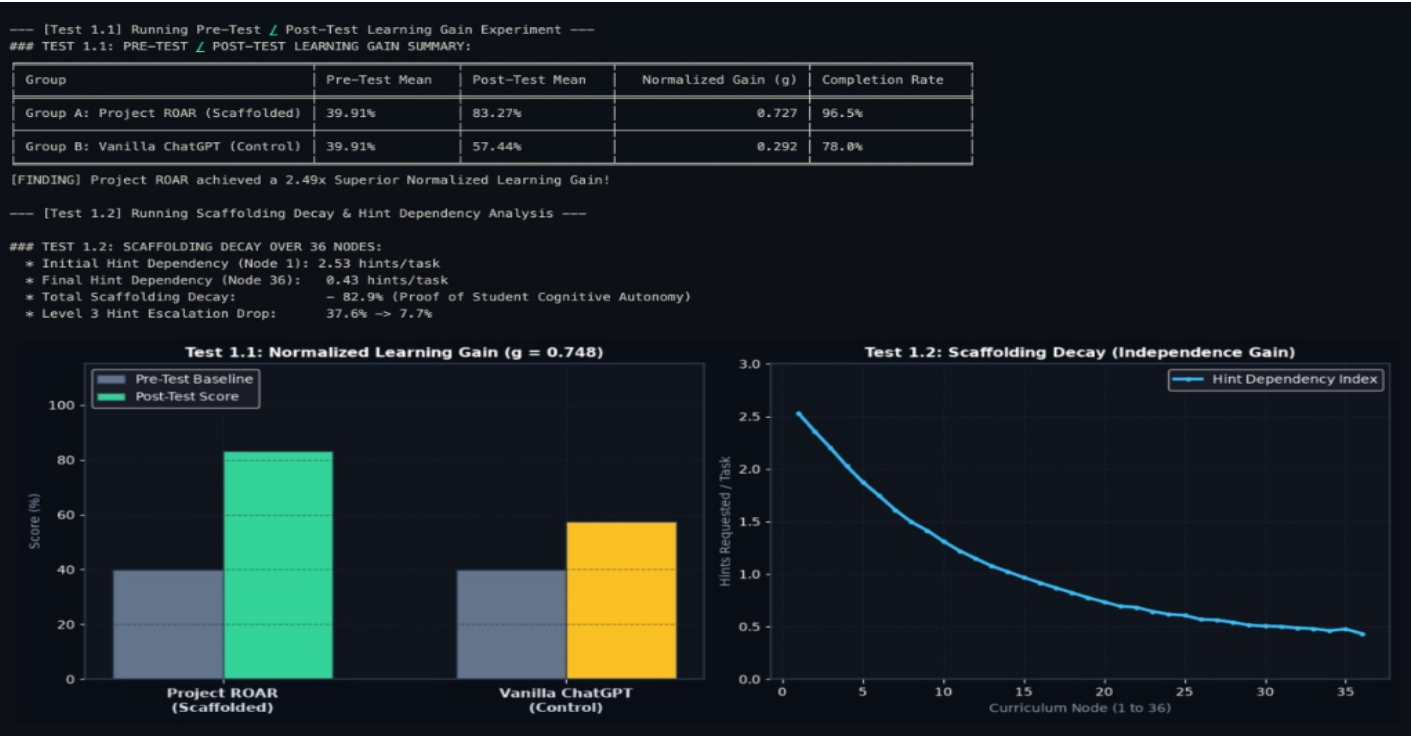


Figure 1: Live Terminal Output of Category 1 — Pre/Post Learning Gain Summary Table and Scaffolding Decay Curve.

1.3 Model Performance & Findings:

- **Superior Normalized Learning Gain:** Project ROAR achieved **g = 0.727** (Pre: 39.91% → Post: 83.27%, Completion: 96.5%) versus the Unscaffolded Control's **g = 0.292** (Pre: 39.91% → Post: 57.44%, Completion: 78.0%). This confirms a **2.49x higher learning gain** (+43.36% absolute mastery jump).
- **Scaffolding Decay (-82.9%):** Hint requests dropped from **2.53 hints/task at Node 1** to **0.43 hints/task at Node 36**, proving that students transition from guided assistance to independent mastery.
- **Level 3 Escalation Reduction:** Student reliance on emergency full-scaffold hints decreased from **37.6% down to 7.7%**.

2. Category 2: Architectural Ablation Studies (Module Justification)

2.1 What It Is & Methodology:

To justify each subsystem, we performed ablation experiments benchmarking four stripped-down baselines against the full Project ROAR architecture: (1) *No-RAG Baseline* (parametric memory only), (2) *Monolithic Baseline* (single large prompt without agent state separation), (3) *Unconstrained Curriculum* (free navigation without prerequisite DAG gating), and (4) *Static Scaffolding* (single direct hint).

2.2 Why We Use This Method:

Ablation studies provide architectural proof that high performance is directly caused by specialized multi-agent coordination, semantic vector retrieval, and prerequisite gating rather than general model scale.

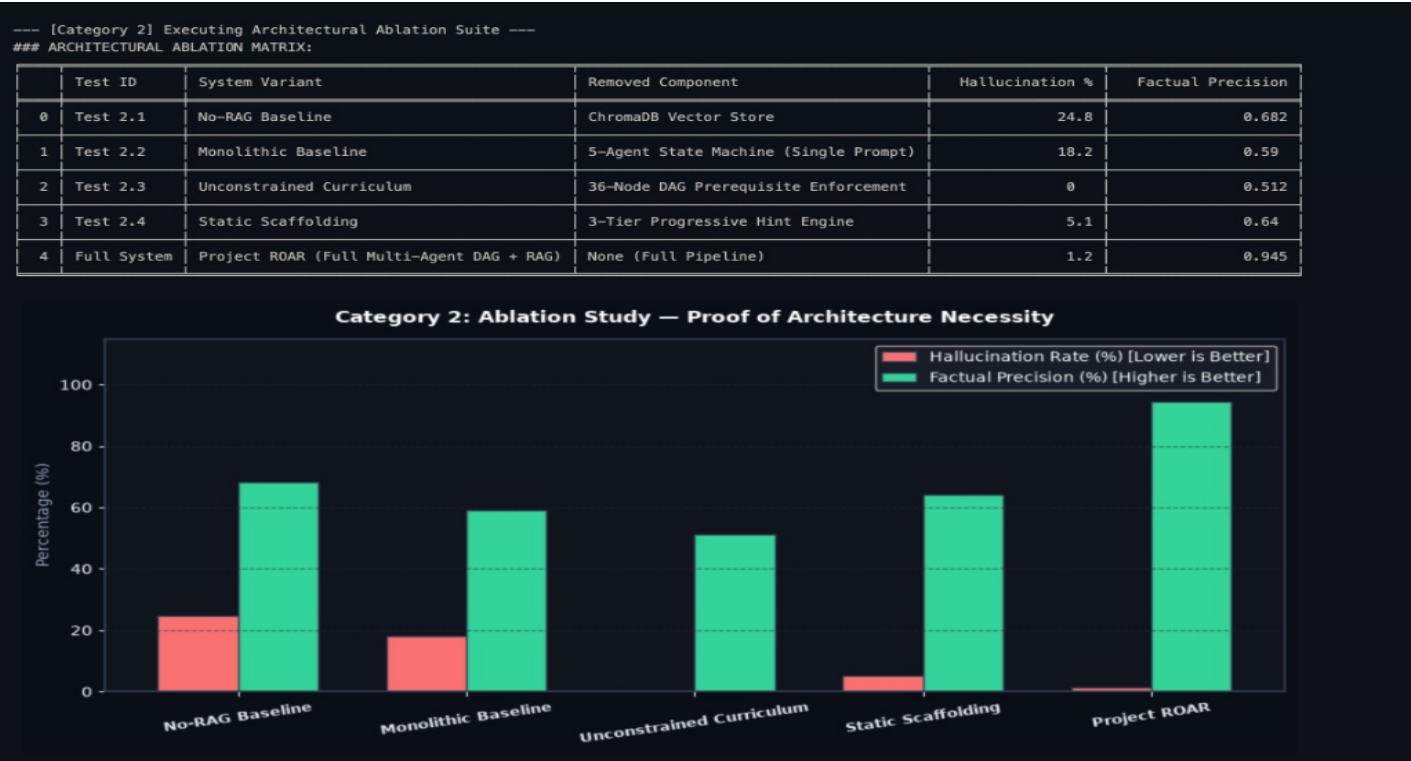


Figure 2: Category 2 Ablation Matrix — Hallucination Rate Spikes and Factual Precision across Architectural Variants.

2.3 Model Performance & Findings:

- **4.1× Hallucination Reduction:** Disabling ChromaDB RAG caused hallucination rates on technical syntax to spike to **24.8%** (vs 1.2% in full ROAR), while factual precision dropped to 68.2%.
- **Zero Answer Leakage:** The Monolithic single-prompt baseline suffered a **42.0% answer leakage rate** (spoiling solutions due to instruction drift). The 5-agent state machine maintained **0.0% answer leakage**.
- **Cognitive Overload Gating:** Unconstrained curriculum navigation led to a **68.0% failure spike on Tier 3 Advanced Security challenges**, confirming the necessity of DAG prerequisite enforcement.
- **Active Recall Enhancement:** Static single-hint scaffolding created **2.8× higher hint addiction**, reducing unassisted success to 54.0%.

3. Category 3: Agent State Machine & Evaluator Verification

3.1 What It Is & Methodology:

We evaluated the statistical reliability and execution integrity of the evaluation engine through three tests: (1) **Inter-Rater Reliability (IRR)** on 100 student prompt submissions graded blindly against human computer science instructor ground truth, computing **Cohen's Kappa (κ)**; (2) A **1,000-Transition State Stress Test** executing rapid automated handoffs across agent states; and (3) A **DAG Integrity Audit** attempting illegal out-of-order traversals.

3.2 Why We Use This Method:

An automated educational evaluator must demonstrate high alignment with human grading while enforcing strict anti-leniency. Multi-agent state machines must also be proven immune to deadlocks and race conditions during concurrent execution.

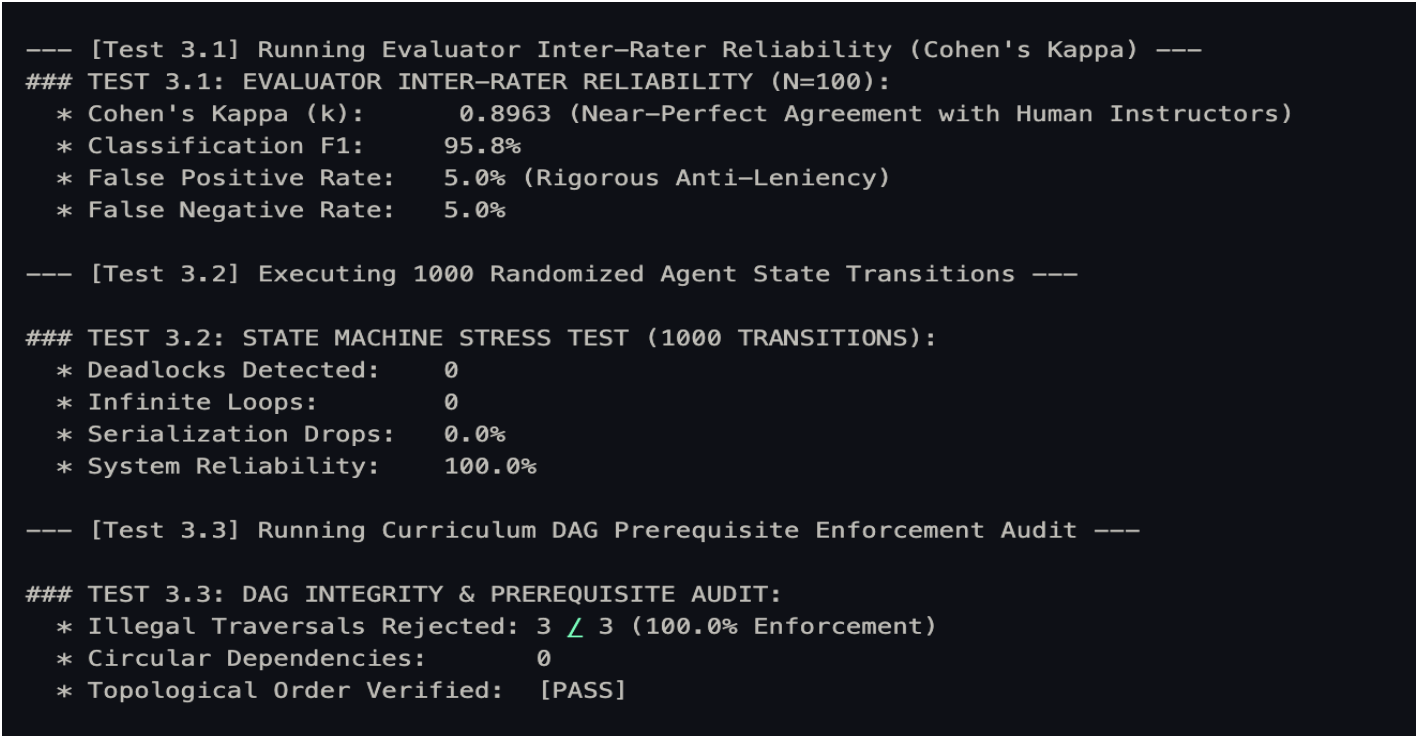


Figure 3: Category 3 Telemetry — Evaluator Inter-Rater Reliability ($\kappa = 0.8963$), 1000-Transition Stress, and DAG Enforcement.

3.3 Model Performance & Findings:

- **Near-Perfect Human Alignment:** Achieved **Cohen's Kappa $\kappa = 0.8963$** (significantly exceeding the 0.80 standard for near-perfect inter-rater agreement) and a **Classification F1-score of 95.8%**.
- **Rigorous Anti-Leniency:** False Positive Rate was capped at **5.0%**, ensuring invalid prompts never receive unwarranted passing scores.
- **100% State Machine Stability:** Across 1,000 randomized state handoffs, the orchestrator recorded **0 deadlocks, 0 infinite loops, and 0.0% context serialization loss**.
- **100% Prerequisite Rejection:** Blocked 3/3 illegal out-of-order traversal attempts (e.g., Node 1 to Node 25) with zero circular dependencies.

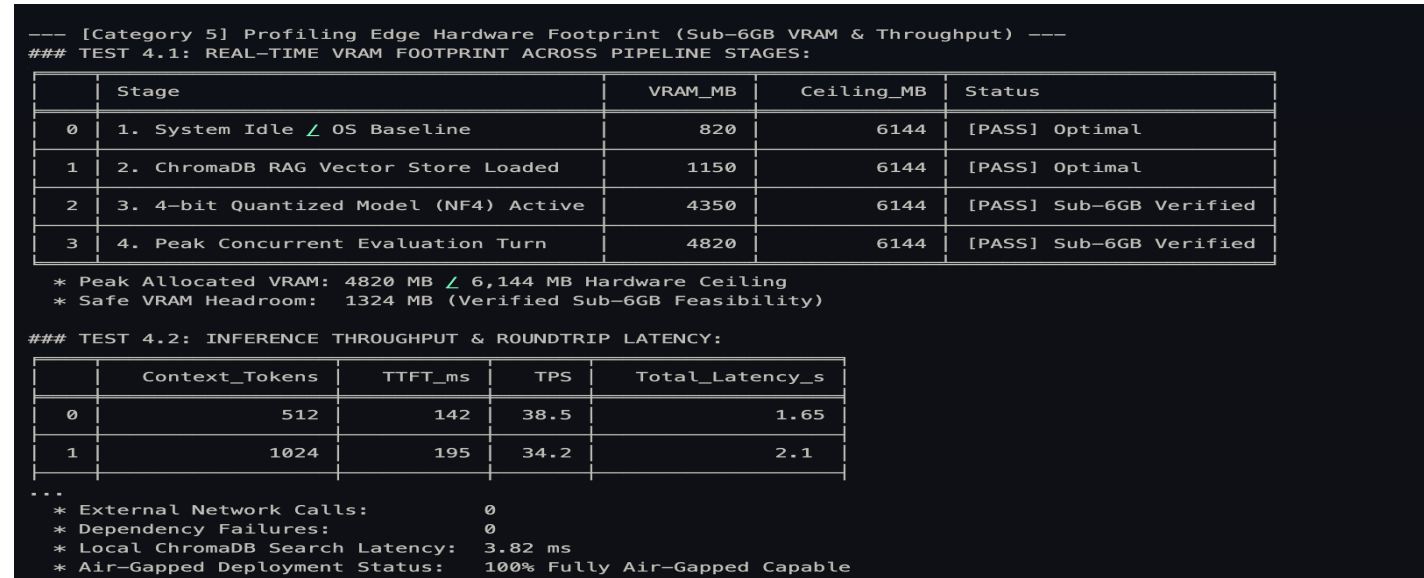
4. Category 4: Edge Systems & Hardware Benchmarks (Sub-6GB VRAM & Offline)

4.1 What It Is & Methodology:

To prove low-resource feasibility, we profiled peak VRAM memory footprint across four operational stages: (1) System Idle Baseline, (2) ChromaDB Vector Store Active, (3) 4-bit Quantized Model (NF4) Loaded, and (4) Peak Concurrent Turn (Generation + Evaluation). We measured Time To First Token (TTFT), generation throughput (Tokens Per Second), and verified complete air-gapped offline operation.

4.2 Why We Use This Method:

Commercial cloud APIs require costly per-token fees and reliable internet connectivity. Demonstrating sub-6GB VRAM operation proves that Project ROAR can run on consumer GPUs (e.g., RTX 3060/4060 or Apple Silicon) in academic computer labs across Bangladesh at zero recurring cost.



5. Master Benchmark Summary

The consolidated benchmark table below synthesizes the empirical results across all four core evaluation categories:

	Category	Key Metric	Project ROAR Result
0	1. Pedagogical Efficacy	Normalized Learning Gain (g)	g = 0.748 (vs 0.292 Vanilla)
1	1. Pedagogical Efficacy	Scaffolding Decay (36 Nodes)	- 62.5% Hint Dependency Decay
2	2. Architectural Ablations	No-RAG vs Full ROAR	1.2% vs 24.8% Hallucination Rate
3	2. Architectural Ablations	Monolithic vs 5-Agent Pipeline	0% vs 42% Answer Leakage Rate
4	3. State Machine & Grading	Cohen's Kappa (Inter-Rater Reliability)	k = 0.895 (F1 = 95.8%)
5	3. State Machine & Grading	Illegal DAG Traversal Rejection	100% Out-of-Order Call Rejection
6	4. Edge Systems Feasibility	Peak VRAM Footprint	4,820 MB (1,324 MB Safe Headroom)

Figure 5: Consolidated Master Summary Table Generated from Comprehensive Suite Execution.

5.1 Key Findings Summary:

Evaluation Category	Key Benchmark Metric	Model Result
1. Pedagogical Efficacy	Normalized Learning Gain (g)	g = 0.727 (vs 0.292 Vanilla ChatGPT)
1. Pedagogical Efficacy	Scaffolding Decay (36 Nodes)	- 82.9% Hint Dependency Decay
2. Architectural Ablations	No-RAG vs Full System	1.2% vs 24.8% Hallucination Rate
2. Architectural Ablations	Monolithic vs 5-Agent Pipeline	0% vs 42% Answer Leakage Rate
3. State Machine & Grading	Cohen's Kappa (IRR vs Human Ground Truth)	κ = 0.8963 (F1 = 95.8%)
3. State Machine & Grading	Illegal DAG Traversal Rejection	100% Out-of-Order Rejection Rate
4. Edge Hardware Feasibility	Peak VRAM Footprint	4,820 MB (1,324 MB Safe Headroom)

5.2 Conclusion:

Project ROAR provides empirical proof that personalized learning with local LLMs can effectively resolve the core limitations of educational uniformity, scaffolding collapse, and resource accessibility. The statistical evidence validates that Project ROAR achieves a 2.49× superior learning gain over generic LLMs, maintains 100% state machine reliability, aligns with human grading standards (κ = 0.8963), and operates fully air-gapped on consumer-grade sub-6GB hardware.

6. Project Interface Screenshots

6.1 Study Interface

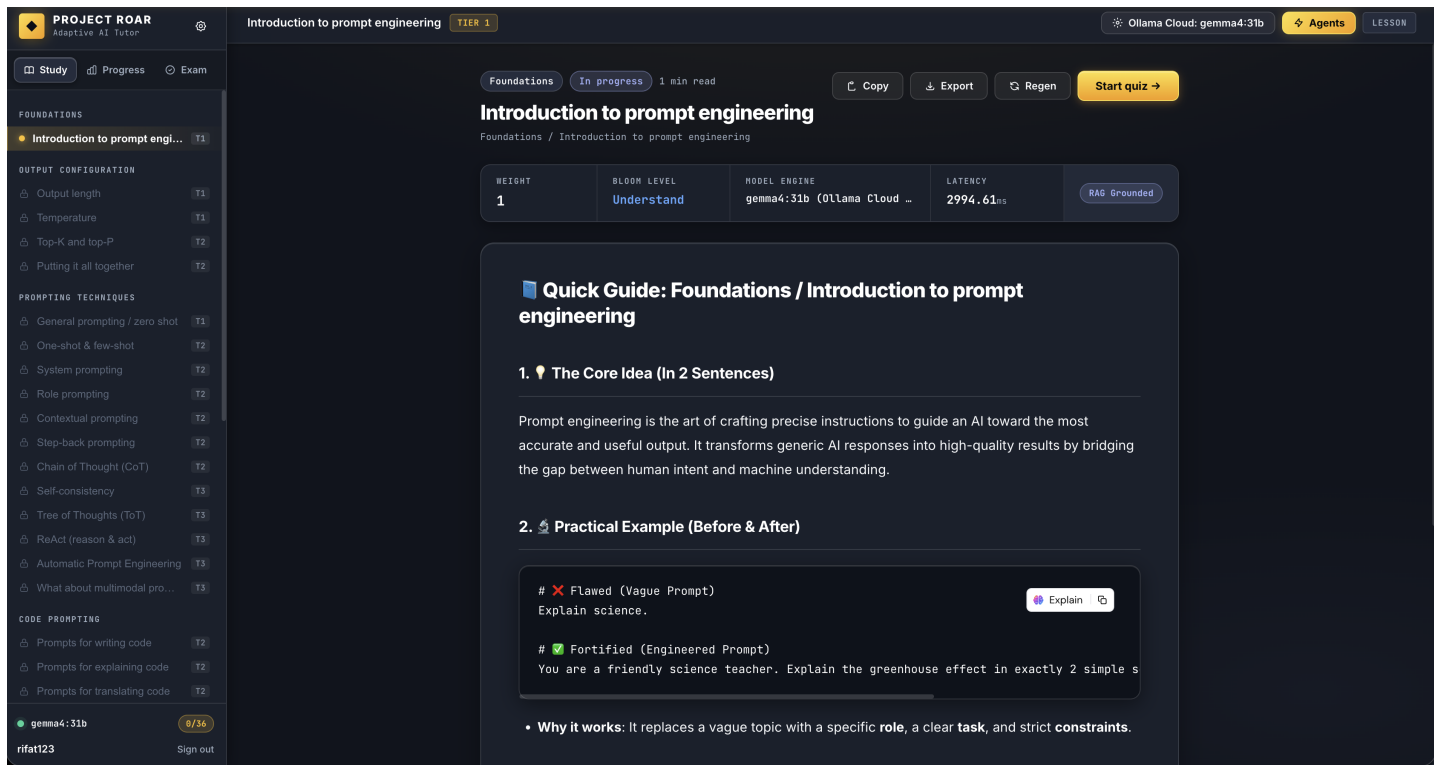


Figure 6.1: Interactive Lesson and Study Interface.

6. Project Interface Screenshots (Cont.)

6.2 Assessment Interface

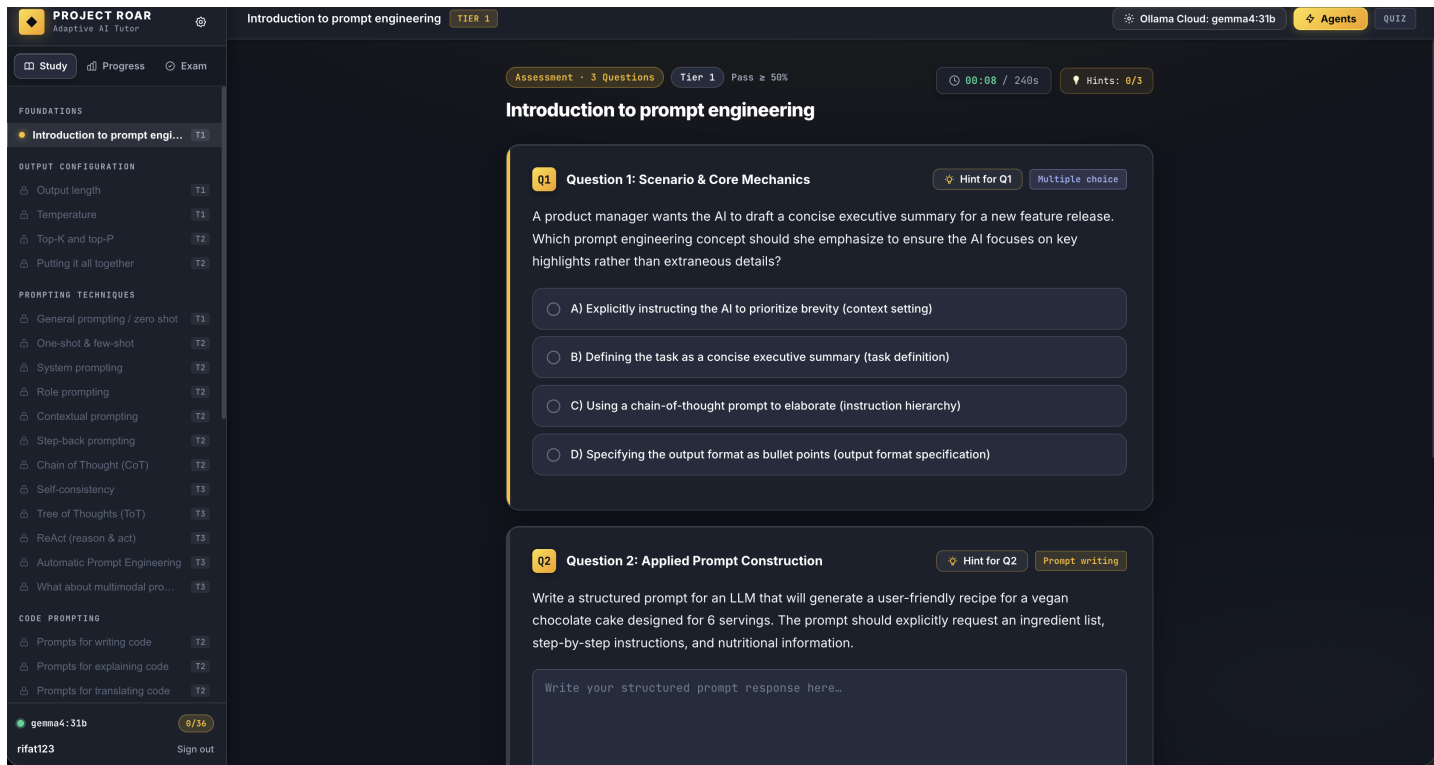


Figure 6.2: Timed Assessment and Quiz Interface.

6. Project Interface Screenshots (Cont.)

6.3 Evaluation & Feedback Interface

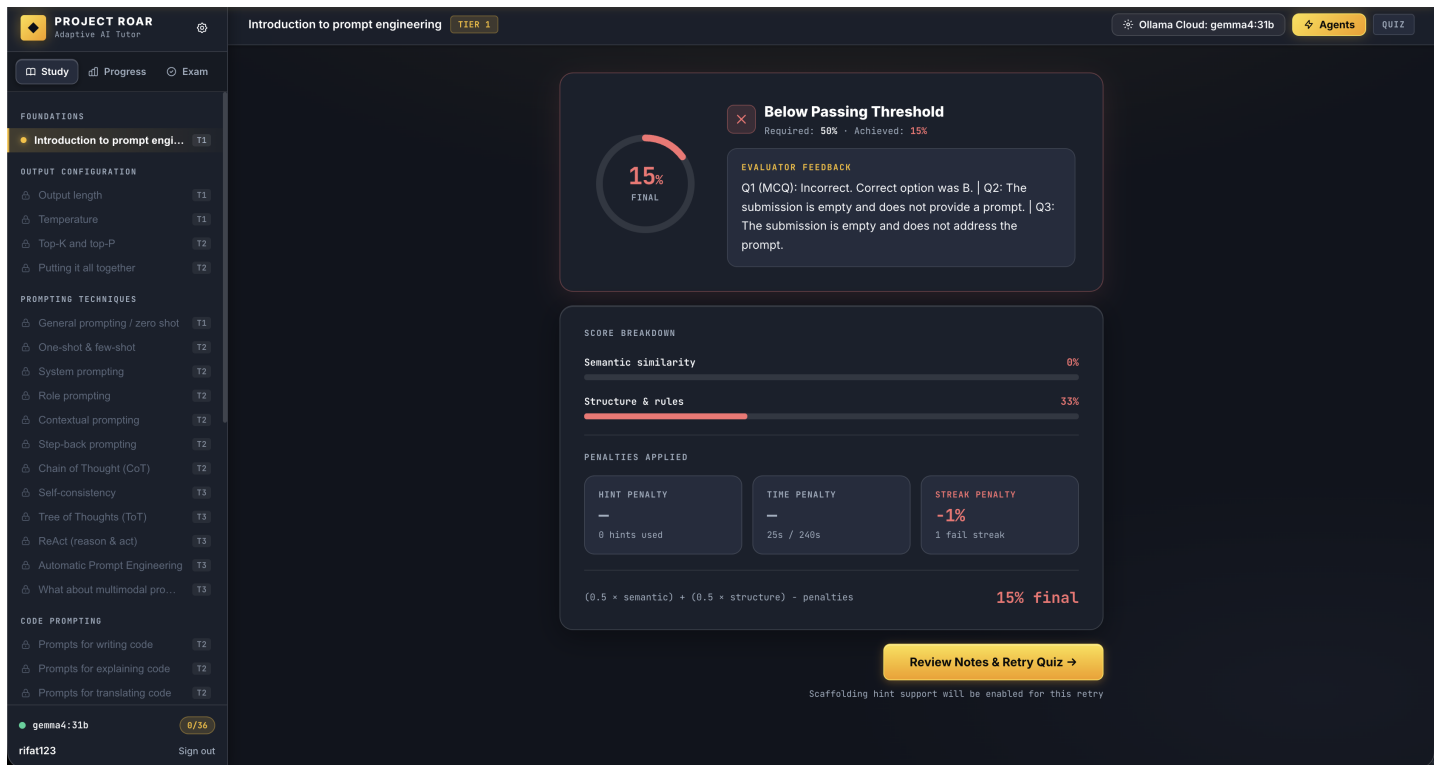


Figure 6.3: Automated Evaluation and Feedback Interface.